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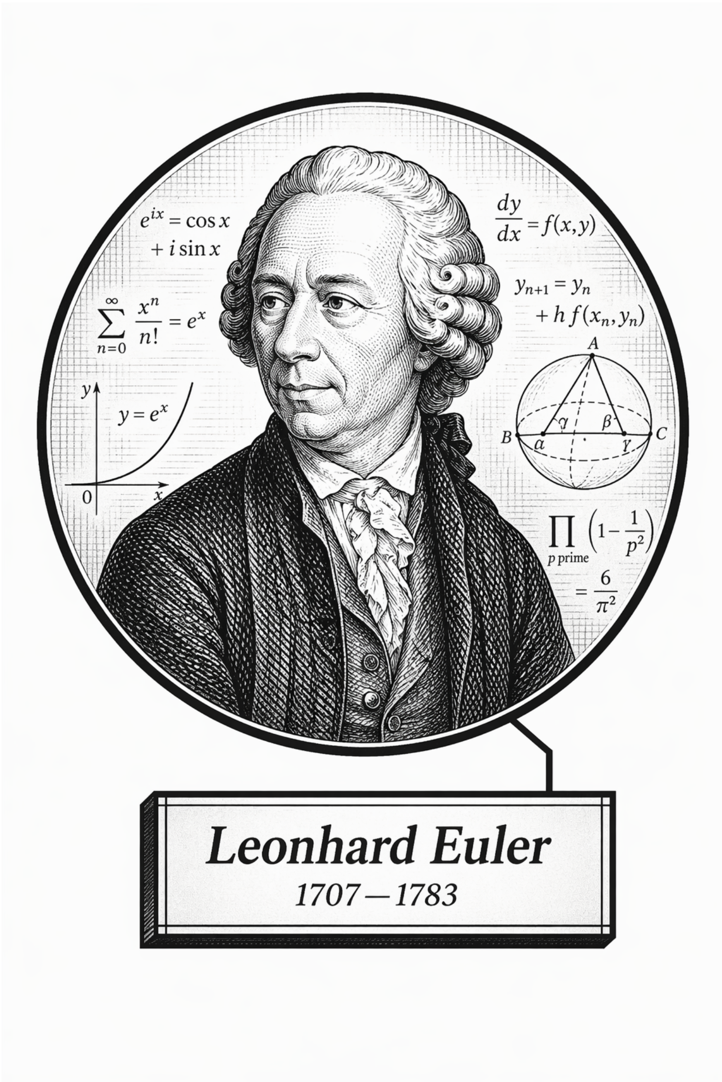

Frontispiece placeholder.
Leonhard Euler, 1707–1783

Volume VI

Computational Mathematics

Self-Study Notes

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Part I

Computational Mathematics

Chapter 1

Calculus

Where You Are in the Journey

Real Numbers $\to$ **Calculus (Computational Fluency)** $\to$ Real Analysis $\to$ Multivariable Analysis $\to$ Differential Equations $\to \cdots$

Volume VI treats calculus as a working toolkit. The real number system is already available; here the emphasis is on computing limits, rates of change, accumulation, series, and multivariable quantities accurately and fluently.

Structural Role: Calculus as Computational Fluency

This chapter is a computational field manual. It records definitions, computational theorem statements, rule tables, examples, method notes, and exercises. Rigorous proofs of the main calculus theorems belong in the Real Analysis volume.

The guiding dependency pattern is:

$$\text{Volume VI computational theorem} \longrightarrow \text{Volume II rigorous theorem} \longrightarrow \text{proof file.}$$

Structural Roadmap

1. Limits: numerical behavior, algebraic methods, one-sided behavior, infinity, squeeze arguments, and asymptotes.
2. Continuity: pointwise and interval continuity, discontinuities, and the Intermediate Value Theorem as a computational tool.
3. Derivatives: difference quotients, rule algebra, standard derivative tables, chain-rule patterns, implicit and logarithmic differentiation, and applications.
4. Integrals: antiderivatives, definite integrals, the Fundamental Theorem of Calculus as a computational bridge, techniques, improper integrals, and applications.
5. Sequences and series: convergence, standard tests, power series, and Taylor/Maclaurin expansions.
6. Multivariable calculus: vectors, curves, partial derivatives, gradients, multiple integrals, line and surface integrals, and the vector-calculus theorems.

1.1 Limits

Definitions

Definition (Intuitive Limit Idea). The limit $\lim_{x\to a} f(x) = L$ says that the values of $f(x)$ can be made close to L by taking x close to a, without requiring $f(a)$ itself to be defined or equal to L.

Definition (Epsilon-Delta Reference Definition). The statement $\lim_{x\to a} f(x) = L$ means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

In Volume VI this definition is mainly a reference point for interpreting limit notation; detailed proofs belong in Real Analysis.

Definition (One-Sided Limits). The right-hand limit $\lim_{x\to a^+} f(x)$ tracks values with $x > a$. The left-hand limit $\lim_{x\to a^-} f(x)$ tracks values with $x < a$. The two-sided limit exists exactly when both one-sided limits exist and agree.

Definition (Limits at Infinity). The notation $\lim_{x\to\infty} f(x) = L$ describes the value approached by $f(x)$ as x becomes arbitrarily large. The notation $\lim_{x\to a} f(x) = \infty$ describes unbounded growth near a.

Definition (Indeterminate Forms). Forms such as $0/0$, ∞/∞, $0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, and ∞^0 do not determine a limit by inspection. They signal that algebraic or analytic rewriting is needed.

Definition (Asymptote). A horizontal, vertical, or oblique asymptote records limiting behavior of a graph near infinity or near a point where the function grows without bound.

Computational Methods

Situation	First method to try
Polynomial or rational expression	Direct substitution, then factoring
Radicals	Rationalize with the conjugate
Common factor $0/0$	Cancel only after factoring
Trigonometric small-angle form	Use standard trigonometric limits
Bounded expression times vanishing expression	Squeeze theorem
Large rational functions	Compare leading powers

Example (Factoring a Removable Form). For

$$\lim_{x\to 3} \frac{x^2 - 9}{x - 3},$$

factor $x^2 - 9 = (x - 3)(x + 3)$, cancel the common nonzero factor for $x \neq 3$, and evaluate $x + 3$ at $x = 3$. The limit is 6.

Theorem Reference

Theorem (Limit Laws). *When the relevant limits exist, sums, differences, scalar multiples, products, and quotients of functions have limits computed by applying the same algebraic operation to the limiting values. Quotients require a nonzero denominator limit.*

Theorem (Squeeze Theorem). *If* $g(x) \leq f(x) \leq h(x)$ *near* a *and*

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L,$$

then $\lim_{x \to a} f(x) = L$.

Limits Exercises

Status: stubbed.

Exercise sets for limits will be routed here.

1.2 Continuity

Definitions

Definition (Continuity at a Point). A function f is continuous at a when $f(a)$ is defined, $\lim_{x\to a} f(x)$ exists, and

$$\lim_{x\to a} f(x) = f(a).$$

Definition (Continuity on an Interval). A function is continuous on an interval when it is continuous at each interior point and has the appropriate one-sided continuity at any endpoint included in the interval.

Definition (Common Discontinuities). A removable discontinuity is a hole, a jump discontinuity has unequal one-sided limiting values, and an infinite discontinui occurs when function values become unbounded near the point.

Computational Methods

Task	Practical check
Polynomial continuity	Continuous everywhere
Rational continuity	Exclude zeros of the denominator
Piecewise continuity	Check each piece and the junction values
Root/log continuity	Check the natural domain
Classify a discontinuity	Compare one-sided behavior and function value

Example (Piecewise Junction). For a piecewise function at $x = a$, compute $f(a)$, $\lim_{x\to a^-} f(x)$, and $\lim_{x\to a^+} f(x)$. Continuity requires all three quantities to exist and agree.

Theorem Reference

Theorem (Algebra of Continuous Functions). *Sums, differences, scalar multiples, products, quotients with nonzero denominator, and compositions of continuous functions are continuous wherever the resulting expressions are defined.*

Theorem (Intermediate Value Theorem). *If f is continuous on $[a, b]$ and N lies between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that $f(c) = N$.*

Continuity Exercises

Status: stubbed.

Exercise sets for continuity will be routed here.

1.3 Derivatives

Definition

Definition (Derivative as a Difference Quotient Limit). The derivative of f at a is

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h},$$

when this limit exists.

Remark (Slope and Rate of Change). Geometrically, $f'(a)$ is the slope of the tangent line to $y = f(x)$ at $x = a$. In applications, it is the instantaneous rate of change of the output with respect to the input.

Example (Power from the Definition). For $f(x) = x^2$,

$$\frac{f(a+h) - f(a)}{h} = \frac{(a+h)^2 - a^2}{h} = 2a + h,$$

so $f'(a) = 2a$. This is a short computational derivation, not a theorem proof vault entry.

Derivative Algebra

Theorem (Constant, Sum, Difference, and Scalar Multiple Rules). *For differentiable functions f and g and a constant c,*

$$(c)' = 0, \qquad (f+g)' = f' + g', \qquad (f-g)' = f' - g', \qquad (cf)' = cf'.$$

Theorem (Product Rule). *If f and g are differentiable at x, then*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Theorem (Quotient Rule). *If f and g are differentiable at x and $g(x) \neq 0$, then*

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}.$$

Theorem (Chain Rule). *If g is differentiable at x and f is differentiable at $g(x)$, then*

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

Standard Derivative Table

Function	Derivative
x^n	nx^{n-1}
$\sqrt{x}$	$1/(2\sqrt{x})$
$1/x$	$-1/x^2$
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\arcsin x$	$1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$

Chain-Rule Patterns

Outer form	Pattern
$(u(x))^n$	$n(u(x))^{n-1}u'(x)$
$e^{u(x)}$	$e^{u(x)}u'(x)$
$\ln(u(x))$	$u'(x)/u(x)$
$\sin(u(x))$	$\cos(u(x))u'(x)$
$\sqrt{u(x)}$	$u'(x)/(2\sqrt{u(x)})$

Remark (Method). When differentiating a composition, name the inside expression u, apply the derivative rule to the outside function, and multiply by u'.

Implicit Differentiation

Remark (Method). Differentiate both sides with respect to x, treating y as a function of x. Every derivative of an expression involving y picks up a factor of dy/dx. Then solve algebraically for dy/dx.

Example (Circle). From $x^2 + y^2 = 1$, differentiate to get

$$2x + 2y\frac{dy}{dx} = 0, \qquad \frac{dy}{dx} = -\frac{x}{y}.$$

Logarithmic Differentiation

Remark (Method). Use logarithmic differentiation when products, quotients, powers with variable exponents, or many repeated factors make direct differentiation unwieldy. Take logarithms, expand using log laws, differentiate implicitly, and multiply by the original function.

Example (Variable Power). If $y = x^x$ for $x > 0$, then

$$\ln y = x \ln x, \qquad \frac{y'}{y} = \ln x + 1, \qquad y' = x^x(\ln x + 1).$$

Applications

Application	Derivative role
Tangent lines	Slope at a point
Linear approximation	$f(a) + f'(a)(x - a)$
Critical points	Solve $f'(x) = 0$ or undefined
Curve sketching	Use f' and f'' sign information
Optimization	Find and compare candidates
Related rates	Differentiate a relation with respect to time

Theorem (Mean Value Theorem). *If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem (L'Hopital's Rule). *For suitable differentiable functions producing an indeterminate form $0/0$ or ∞/∞,*

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

when the derivative quotient limit exists under the hypotheses of the rule.

Exercise-vault records are organized by mutable topic paths. Stable exercise set and item identifiers are canonical and are recorded in `exercise-ledger.yaml`.

Derivatives

Status: active.

Derivative Rules

Exercise Set: Derivative Rules **Stable set ID:** `set-derivative-rules-001`. **Mode:** computation. **Topic:** derivatives. **Status:** memorialized.

1. **Stable item ID:** `ex-deriv-rules-001`.

   Exercise. Find the derivative of

   $$y = 7x^5 - 4x^3 + 9x - 2.$$

   Solution. By the power rule and linearity,

   $$\frac{dy}{dx} = 35x^4 - 12x^2 + 9.$$

   Verification. The result is correct.

2. **Stable item ID:** `ex-deriv-rules-002`.

   Exercise. Find the derivative of
   $$f(x) = x^2 \sin x.$$

   Solution. By the product rule,
   $$f'(x) = 2x \sin x + x^2 \cos x.$$

   Verification. The result is corrected. The problem was initially copied as a quotient, but the memorialized exercise is the product $x^2 \sin x$.

   Rework note. Preserve the correction in the ledger so later refactors do not recover the discarded quotient transcription.

3. **Stable item ID:** `ex-deriv-rules-003`.

   Exercise. Find the derivative of
   $$y = \frac{x^2 + 1}{x - 3}.$$

   Solution. By the quotient rule,
   $$\frac{dy}{dx} = \frac{(x-3)(2x) - (x^2+1)}{(x-3)^2} = \frac{x^2 - 6x - 1}{(x-3)^2}.$$

   Verification. The result is correct, with the final simplification added.

`Mixed Derivative Practice`

`Exercise Set: Mixed Derivative Practice` **Stable set ID:** `set-derivative-mixed-001`. **Mode:** computation. **Topic:** derivatives. **Status:** generated placeholder.

This exercise set is reserved for future mixed derivative practice. No solved exercises have been memorialized here yet.

1.4 Integrals

Definitions

Definition (Antiderivative). An antiderivative of f on an interval is a function F such that $F' = f$ on that interval.

Definition (Definite Integral). The definite integral $\int_a^b f(x)\,dx$ measures signed accumulation of f over $[a, b]$. Computationally, it is evaluated by antiderivatives whenever the Fundamental Theorem of Calculus applies.

Theorem (Fundamental Theorem of Calculus). *If $F' = f$ on an interval containing a and b, then*

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Integral Algebra

Rule	Formula
Linearity	$\int (af + bg) = a \int f + b \int g$
Interval additivity	$\int_a^b f + \int_b^c f = \int_a^c f$
Reversal	$\int_a^b f = -\int_b^a f$
Zero width	$\int_a^a f = 0$
Constant multiple	$\int_a^b c\,dx = c(b - a)$

Techniques

Technique	Use when
Substitution	The integrand contains a composition and its derivative
Integration by parts	The integrand is a product with a simpler derivative
Trigonometric integrals	Powers of trig functions can be reduced
Trigonometric substitution	Radicals match $a^2 - x^2$, $a^2 + x^2$, or $x^2 - a^2$
Partial fractions	A rational function has a factorable denominator
Improper integrals	The interval is infinite or the integrand is unbounded

Remark (Method). Before choosing a technique, simplify algebraically, check for a direct antiderivative, and look for an inside function whose derivative is also present.

Applications

Application	Typical integral
Area	$\int_a^b (\text{top} - \text{bottom})\,dx$
Volume by slices	$\int_a^b A(x)\,dx$
Volume by shells	$\int 2\pi(\text{radius})(\text{height})\,dx$
Arc length	$\int_a^b \sqrt{1 + (y')^2}\,dx$
Surface area	$\int 2\pi(\text{radius})\,ds$

Integrals Exercises

Status: stubbed.

Exercise sets for integrals will be routed here.

1.5 Sequences and Series

Sequences

Definition (Sequence). A sequence is a function from positive integers to numbers, usually written (a_n).

Definition (Convergence of a Sequence). The sequence (a_n) converges to L when a_n approaches L as $n \to \infty$. Computationally, this is evaluated using algebraic simplification, standard limits, squeezing, monotonicity, and comparison.

Series

Definition (Series and Partial Sums). The series $\sum_{n=1}^{\infty} a_n$ is studied through its partial sums

$$s_N = \sum_{n=1}^{N} a_n.$$

The series converges when the sequence (s_N) converges.

Series	Convergence pattern
Geometric $\sum ar^n$	Converges for $\lvert r\rvert < 1$
p-series $\sum 1/n^p$	Converges for $p > 1$

Convergence Tests

Test	Best used when
Term test	Terms do not tend to zero
Comparison	Positive terms resemble a known benchmark
Limit comparison	Ratios to a benchmark have a finite positive limit
Integral test	Terms come from a positive decreasing function
Alternating series test	Signs alternate and magnitudes decrease to zero
Ratio test	Factorials or exponentials appear
Root test	nth powers appear

Power Series

Definition (Power Series). A power series centered at a has the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n.$$

Its convergence is organized by its radius and interval of convergence.

Definition (Taylor and Maclaurin Series). The Taylor series of f centered at a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The Maclaurin series is the special case $a = 0$.

Sequences and Series Exercises

Status: stubbed.

Exercise sets for sequences and series will be routed here.

1.6 Multivariable Calculus

Vectors and Curves

Definition (Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$). Vectors encode magnitude and direction. In $\mathbb{R}^2$ and $\mathbb{R}^3$, they are manipulated through component arithmetic, dot products, cross products, projections, and parametrized lines and planes.

Definition (Vector-Valued Function). A vector-valued function $\mathbf{r}(t)$ traces a curve. Its derivative $\mathbf{r}'(t)$ is the velocity vector and points in the tangent direction when nonzero.

Partial Derivatives

Definition (Partial Derivative). For a function $f(x, y)$, the partial derivative f_x differentiates with respect to x while holding y constant. The derivative f_y does the same with respect to y.

Remark (Several-Variable Chain Rule). The chain rule in several variables tracks every path by which a variable can affect the output. In computations, draw the dependency diagram before writing the derivative formula.

Definition (Tangent Plane). For $z = f(x, y)$, the tangent plane at (a, b) is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

when the required differentiability hypotheses hold.

Gradients and Directional Derivatives

Definition (Gradient). The gradient of $f(x, y, z)$ is

$$\nabla f = \langle f_x, f_y, f_z \rangle.$$

It points in the direction of steepest increase.

Definition (Directional Derivative). For a unit vector $\mathbf{u}$,

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.$$

Remark (Lagrange Multipliers). To optimize f subject to a constraint $g = c$, solve

$$\nabla f = \lambda \nabla g, \qquad g = c,$$

then compare candidate values.

Multiple Integrals

Definition (Double and Triple Integrals). Double and triple integrals accumulate a quantity over a region in the plane or in space:

$$\iint_R f\, dA, \qquad \iiint_E f\, dV.$$

Coordinate system	Common use
Polar	Circular regions in the plane
Cylindrical	Circular symmetry around an axis
Spherical	Radial symmetry around a point

Vector Calculus Theorems

Definition (Line and Surface Integrals). Line integrals accumulate along curves. Surface integrals accumulate over surfaces, often measuring flux through the surface.

Theorem (Green's Theorem). *For suitable planar regions and vector fields, a circulation integral around the boundary equals a double integral of curl over the region.*

Theorem (Stokes' Theorem). *For suitable oriented surfaces, a circulation integral around the boundary equals a surface integral of curl over the surface.*

Theorem (Divergence Theorem). *For suitable solid regions, the flux across the boundary equals the triple integral of divergence over the region.*

Multivariable Calculus Exercises

Status: stubbed.

Exercise sets for multivariable calculus will be routed here.

1.7 Capstone

To be populated.

Chapter 2

Geometric Modeling

Where You Are in the Journey

Analytical Geometry $\to$ Linear Algebra $\to$ **Geometric Modeling** $\to$ Differential Geometry $\to \cdots$

How we got here. Analytical geometry gave algebraic descriptions of geometric objects in coordinates. Linear algebra provided the language of transformations. Geometric modeling applies both to represent and manipulate *complex* geometric shapes — curves and surfaces that cannot be described by a single equation — using piecewise polynomial constructions.

What this chapter builds. Bézier curves and their de Casteljau algorithm; B-splines and NURBS; parametric surface patches; subdivision surfaces; affine and projective transformations in homogeneous coordinates; the relationship between geometric modeling constructions and the underlying algebraic and differential geometry; and curvature analysis of modeled curves and surfaces.

Where this leads. Geometric modeling is the computational face of differential geometry. The smooth curves and surfaces studied abstractly in differential geometry are approximated and manipulated concretely here. For the main curriculum trajectory, this chapter provides applied grounding and a source of concrete examples before the more abstract machinery of manifolds and Riemannian geometry.

Structural Role: Geometric Modeling Geometry Made Computable

Geometric modeling is where the abstract geometry sequence meets computation. Bézier curves, B-splines, and NURBS are not merely engineering tools: they are polynomial and rational maps from parameter space into $\mathbb{R}^n$, and every property of interest (smoothness, curvature, convex hull containment) is a theorem in disguise. The de Casteljau algorithm is a recursive application of affine interpolation; subdivision surfaces are limit constructions whose analysis uses the spectral theory of subdivision matrices.

This chapter serves two roles in the project. First, it provides a concrete, computable bridge between the algebra of polynomials and the geometry of curves and surfaces. Second, it is an optional applied detour that reinforces the differential-geometric concepts (parametrisation, arc length, curvature) in a setting where explicit computations are feasible.

Structural Roadmap

Definitions $\longrightarrow$ Main Theorems $\longrightarrow$ Consequences and Structural Insight

The global progression is:

1. Parametric curves: the modeling perspective
2. Bézier curves: Bernstein polynomials and the de Casteljau algorithm

3. Properties of Bézier curves: convex hull, variation diminishing, affine invariance
4. B-splines: knot vectors, basis functions, and local support
5. NURBS: rational extensions and conic sections
6. Parametric surface patches: tensor-product and triangular
7. Continuity conditions: C^k and G^k joins
8. Subdivision curves and surfaces
9. Curvature analysis of parametric curves and surfaces

Remark (Primary sources). Farin, *Curves and Surfaces for CAGD*; Gallier, *Geometric Methods and Applications*; Prautzsch, Boehm & Paluszny, *Bézier and B-Spline Techniques*.

2.1 Parametric Curves and Surfaces

Parametric Curves and Surfaces — Quick Reference

Curve	$C(u) : [a,b] \to \mathbb{R}^n$; vector-valued, one parameter.
Velocity	$C'(u)$; tangent direction and speed of traversal.
Unit tangent	$\hat{T}(u) = C'(u)/\|C'(u)\|$ at a regular point.
Surface	$S(u,v) : R \to \mathbb{R}^3$; vector-valued, two parameters.
Partial derivatives	$S_u = \partial S/\partial u$, $S_v = \partial S/\partial v$; tangent to isoparametric lines.
Unit normal	$N = (S_u \times S_v)/\|S_u \times S_v\|$ at a regular point.
Tangent plane	Spanned by S_u and S_v at the base point.
Directional deriv.	$D_{\mathbf{v}}S = \nabla S \cdot \hat{\mathbf{v}}$; rate of change in direction $\mathbf{v}$.

2.1.1 Parametric Curves

Definition (Parametric Curve)

A *parametric curve* in $\mathbb{R}^n$ is a vector-valued function

$$C : [a,b] \to \mathbb{R}^n, \qquad C(u) = \big(x_1(u),\, x_2(u),\, \ldots,\, x_n(u)\big),$$

where each coordinate function $x_i(u)$ is a real-valued function of the scalar parameter u. The interval $[a,b]$ is typically normalised to $[0,1]$.

Definition (Velocity, Regularity, Unit Tangent)

The *velocity vector* (or *tangent vector*) at u is the derivative

$$C'(u) = \big(x_1'(u),\, x_2'(u),\, \ldots,\, x_n'(u)\big).$$

The curve is *regular* at u if $C'(u) \neq \mathbf{0}$. At a regular point, the *unit tangent vector* is

$$\hat{T}(u) = \frac{C'(u)}{\|C'(u)\|}.$$

The scalar $\|C'(u)\|$ is the *speed* of traversal.

Remark (Properties of parametric curves). Parametric curves carry structure that implicit curves do not:

- **Direction.** The parameter increases from a to b, giving the curve a natural orientation. Reversing the parametrisation reverses the direction.
- **Velocity and speed.** $C'(u)$ measures both the direction of motion and how fast the curve is being traversed. A curve can be *reparametrised* (change of variable $u = \phi(t)$) without changing its shape; the velocity changes but the trace does not.
- **Endpoint interpolation.** The start and end points are $C(a)$ and $C(b)$, accessible directly.
- **Boundedness.** The curve is defined on a closed interval, so it represents a bounded segment, not an infinite line.
- **Cusps.** Where $C'(u) = \mathbf{0}$, the curve may have a cusp (velocity drops to zero, direction may reverse). These are degenerate points excluded by the regularity condition.

2.1.2 Parametric Surfaces

Definition (Parametric Surface)

A *parametric surface* in $\mathbb{R}^3$ is a vector-valued function of two parameters,

$$S : R \longrightarrow \mathbb{R}^3, \qquad S(u,v) = \big(x(u,v),\ y(u,v),\ z(u,v)\big),$$

where $R \subseteq \mathbb{R}^2$ is the *parameter domain* (typically the unit square $[0,1]^2$).

2.1.3 Partial Derivatives and Isoparametric Curves

Definition (Partial Derivative Vectors)

The *partial derivative vectors* of S at (u, v) are

$$S_u(u, v) = \frac{\partial S}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right), \qquad S_v(u, v) = \frac{\partial S}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right).$$

S_u is the tangent vector to the surface in the u-direction: it is the velocity of the curve obtained by fixing v and varying u. S_v is the corresponding tangent vector in the v-direction.

Definition (Isoparametric Curves)

Fixing one parameter on $S(u, v)$ produces a curve on the surface:

- *u-curve* at $v = v_0$: $\quad C_{v_0}(u) = S(u,\, v_0), \quad \text{tangent} = S_u(u, v_0)$.
- *v-curve* at $u = u_0$: $\quad C_{u_0}(v) = S(u_0,\, v), \quad \text{tangent} = S_v(u_0, v)$.

These *isoparametric curves* (isocurves) sweep the surface and form its parameter grid. The two families intersect at every surface point $S(u_0, v_0)$.

2.1.4 Directional Derivative

Definition (Directional Derivative on a Surface)

The *directional derivative* of S at (u_0, v_0) in the direction $\mathbf{d} = (d_u, d_v)$ (a unit vector in parameter space) is

$$D_{\mathbf{d}}\, S = d_u\, S_u(u_0, v_0) + d_v\, S_v(u_0, v_0).$$

It measures the rate of change of position on the surface as the parameter moves in direction $\mathbf{d}$ in the uv-plane. Every tangent vector to the surface at (u_0, v_0) is a directional derivative for some choice of $\mathbf{d}$.

2.1.5 Unit Normal and Tangent Plane

Definition (Regularity, Unit Normal, Tangent Plane)

The surface is *regular* at (u_0, v_0) if $S_u \times S_v \neq \mathbf{0}$ there.
At a regular point, the *unit normal vector* is

$$N(u_0, v_0) = \frac{S_u(u_0, v_0) \;\times\; S_v(u_0, v_0)}{\|S_u(u_0, v_0) \;\times\; S_v(u_0, v_0)\|}\,.$$

The *tangent plane* at $S(u_0, v_0)$ is the plane through $S(u_0, v_0)$ spanned by S_u and S_v:

$$\Pi \;=\; \big\{\, S(u_0, v_0) + \alpha\, S_u + \beta\, S_v \;\big|\; \alpha, \beta \in \mathbb{R} \,\big\}.$$

Its equation in terms of position $\mathbf{p}$ is

$$N \cdot \big(\mathbf{p} - S(u_0, v_0)\big) = 0.$$

Remark (Geometric meaning of $S_u \times S_v$). The magnitude $\|S_u \times S_v\|$ equals the area of the parallelogram spanned by S_u and S_v, which is the *local area element* of the surface. Where $\|S_u \times S_v\| = 0$ the surface is degenerate (the two tangent directions are parallel or one vanishes), and the normal is not defined.

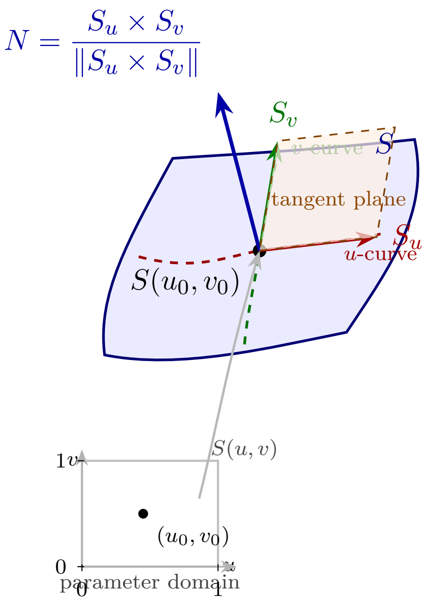

Figure: A parametric surface patch $S(u, v)$ with its parameter domain (lower left). At the point $S(u_0, v_0)$, the partial derivative vectors S_u (red) and S_v (green) are tangent to the isoparametric curves; together they span the tangent plane (orange). The unit normal $N = S_u \times S_v \,/\, \|S_u \times S_v\|$ (blue) is perpendicular to the tangent plane.

Status: Active

This section is actively expanding alongside work in Piegl & Tiller, *The NURBS Book*. Subsequent sections on Bézier curves, B-splines, NURBS, and continuity conditions follow.

Proofs

To be populated.

Capstone

To be populated.

Chapter 3

Computational Geometry

Where You Are in the Journey

Linear Algebra → Geometric Modeling → **Computational Geometry** → Optimization → ⋯

How we got here. Linear algebra supplied coordinates, vectors, dot products, and linear functionals. Geometric modeling showed how finite control data can determine geometric objects. Computational geometry keeps the finite data visible and asks which geometric structures can be defined and constructed from it.

What this chapter builds. The chapter begins with convex hulls: the smallest convex set containing a finite point set, its extreme points, supporting lines, monotone hull structure, and the greedy construction of the upper hull.

Where this leads. Convex hulls connect finite geometry with order, maximization, separation, and algorithmic construction. Later computational-geometry topics use the same pattern: define the geometric object exactly, identify its certificates, and then turn those certificates into algorithms.

3.1 Convex Hulls

Toolkit: Convex Hulls

Ambient space	Finite point sets in $\mathbb{R}^2$.
Convexity	Line segments and convex combinations.
Certificates	Extreme points, supporting lines, and maximizing directions.
Order data	Distinct x-coordinates and slopes between ordered points.
Algorithmic shape	Build the upper hull by repeatedly choosing the next maximal slope.

Remark (Exposition). Computational geometry studies finite geometric data, such as point sets in the plane, using exact definitions and algorithms. Convex hulls are the first organizing example: they connect geometry, order and maximizatio supporting lines, and algorithmic construction.

Convexity and Hulls

Convexity and Convex Hulls

This formulation makes precise the geometric idea of filling in all straight segments forced by a point set. Convex combinations give an algebraic language for the same operation, and the convex hull records the smallest set closed under that operation.

Definition 3.1.1 (Convex Set). A set $S \subseteq \mathbb{R}^2$ is *convex* if for every pair of points $a, b \in S$ and every $\lambda \in [0,1]$,

$$\lambda a + (1-\lambda)b \in S.$$

Equivalently, S contains the line segment between any two of its points.

Definition 3.1.2 (Convex Combination). A *convex combination* of points $q_1, \dots, q_n \in \mathbb{R}^2$ is a point of the form

$$\sum_{i=1}^{n} \lambda_i q_i, \qquad \lambda_i \geq 0 \text{ for all } i, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

The numbers λ_i are called the weights.

Definition 3.1.3 (Convex Hull). The *convex hull* of a set $P \subseteq \mathbb{R}^2$, written $\operatorname{conv}(P)$, is the set of all convex combinations of finitely many points of P:

$$\operatorname{conv}(P) = \left\{ \sum_{i=1}^{n} \lambda_i q_i \;\middle|\; n \geq 1,\;\; q_i \in P,\;\; \lambda_i \geq 0,\;\; \sum_{i=1}^{n} \lambda_i = 1 \right\}.$$

Equivalently, $\operatorname{conv}(P)$ is the smallest convex set containing P, i.e. the intersection of all convex sets that contain P.

Lemma 3.1.4 (Convex hull as the smallest convex set). *Let $P \subseteq \mathbb{R}^2$. Then $\mathrm{conv}(P)$ is convex, contains P, and is contained in every convex set that contains P. Hence*

$$\mathrm{conv}(P) = \bigcap\{K \subseteq \mathbb{R}^2 : K \textit{ is convex and } P \subseteq K\}.$$

Remark (Proof). See the proof stub.

Remark (Dependencies).

- Convex Set
- Convex Combination
- Convex Hull

Supporting Functionals

Supporting Functionals and Vertices

This topic isolates the passage from geometry to optimization. A supporting line is a geometric certificate that a convex set lies on one side of a boundary, while a linear functional turns that certificate into a maximum.

Definition 3.1.5 (Extreme Point). Let $S \subseteq \mathbb{R}^2$ be convex. A point $p \in S$ is an *extreme point* of S if whenever

$$p = \lambda a + (1-\lambda)b, \qquad a, b \in S, \quad \lambda \in (0,1),$$

it follows that $a = b = p$. For a finite point set P, the extreme points of $\mathrm{conv}(P)$ are called the vertices of the convex hull.

Definition 3.1.6 (Supporting Line). A line $\ell \subseteq \mathbb{R}^2$ is a *supporting line* of a convex set S at a point $p \in S$ if $p \in \ell$ and S lies entirely in one of the two closed half-planes bounded by ℓ. Equivalently, there exists $d \in \mathbb{R}^2 \setminus \{0\}$ such that

$$\ell = \{x \in \mathbb{R}^2 : \langle d, x\rangle = \langle d, p\rangle\}$$

and

$$\langle d, q\rangle \leq \langle d, p\rangle \qquad \text{for all } q \in S.$$

Lemma 3.1.7 (Linear functionals preserve convex combinations). *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \ldots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z\rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i\rangle.$$

Consequently, if $\langle d, q_i\rangle \leq M$ for every i, then

$$\langle d, z\rangle \leq M.$$

Remark (Proof). See the proof stub.

Theorem 3.1.8 (Supporting functional characterization of hull vertices). *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of $\operatorname{conv}(P)$ if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p\rangle > \langle d, q\rangle \qquad \textit{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q\rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Remark (Proof). See the proof stub.

Remark (Finite maximum form). The maximum is used here because P is finite. This is the finite-set version of the same supporting-functional intuition that appears in supremum language for more general sets.

Remark (Dependencies).

- Convex Combination
- Convex Hull
- Extreme Point
- Supporting Line
- Linear Functionals Preserve Convex Combinations

Upper Hull Structure

Monotone Slopes and Greedy Construction

This topic converts the supporting-line description into an ordered planar construction. Under the general-position assumptions, each relevant maximum is unique, and the upper hull can be recognized by the strict monotonicity of successive slopes.

Assumption 3.1.9 (General Position for the Hull Theorems). Let $P \subset \mathbb{R}^2$ be a finite set in general position: the points have distinct x-coordinates and no three points are collinear. For $p \in \mathbb{R}^2$, write $x(p)$ and $y(p)$ for its coordinates.

Theorem 3.1.10 (Monotone slope characterization of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-

coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ *satisfy*

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k *is a vertex of*

$$\operatorname{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Remark (Proof). See the proof stub.

Theorem 3.1.11 (Greedy slope construction of the upper hull). *Let* $P \subset \mathbb{R}^2$ *be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k *with*

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \mathop{\arg\max}_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P*, and the slopes encountered along the way are strictly decreasing.*

Remark (Proof). See the proof stub.

Remark (Uniqueness of choices). The distinct x-coordinates make the initial and terminal x-extreme points unique. The no-three-collinear condition makes the slope maximizer in the greedy step unique.

Remark (Proof roadmap). The proof ladder is:

convex combinations $\to$ linear functionals $\to$ supporting lines $\to$ monotone slopes $\to$ greec

The first two lemmas are algebraic. The monotone slope theorem is an analytic-geometry statement about three ordered points. The greedy construction theorem is the algorithmic consequence.

Remark (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices
- TODO: finite set maximum
- TODO: slope of a line

Proofs

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Theorem (Convex hull as the smallest convex set). *Let* $P \subseteq \mathbb{R}^2$. *Then* $\mathrm{conv}(P)$ *is convex, contains* P*, and is contained in every convex set that contains* P*. Hence*

$$\mathrm{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \textit{ is convex and } P \subseteq K\}.$$

Proof. **Professional Standard Proof.**
TODO: Show that finite convex combinations are closed under taking convex combinations and that every convex set containing P contains all finite convex combinations of points of P. □

Proof. **Detailed Learning Proof.**
TODO: Expand the professional proof into steps: containment of P, convexity of $\mathrm{conv}(P)$, containment in any convex superset of P, and equality with the displayed intersection. □

Remark (Proof structure). The proof is a direct unpacking of convex combinations and convexity.

Remark (Dependencies).

- Convex Set
- Convex Combination
- Convex Hull

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Theorem (Linear functionals preserve convex combinations). *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \dots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z \rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i \rangle.$$

Consequently, if $\langle d, q_i \rangle \leq M$ for every i, then

$$\langle d, z \rangle \leq M.$$

Proof. **Professional Standard Proof.**
TODO: Use linearity of the dot product in the second variable and the nonnegative weights summing to 1. □

Proof. **Detailed Learning Proof.**
TODO: Expand the computation of $\langle d, \sum_i \lambda_i q_i \rangle$, then derive the inequality from $\langle d, q_i \rangle \leq M$ and $\sum_i \lambda_i = 1$. □

Remark (Proof structure). The proof is algebraic: distribute the linear functional across the finite sum, then use the convex-combination weights.

Remark (Dependencies).

- Convex Combination
- TODO: dot product / linear functional

Remark (Return). Return to Theorem

Remark (Proof vault). Handwritten proof record

Theorem (Supporting functional characterization of hull vertices). *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of* $\mathrm{conv}(P)$ *if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p\rangle > \langle d, q\rangle \qquad \text{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q\rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Proof. **Professional Standard Proof.**
TODO: Prove one direction by using a unique maximizing functional to create a supporting line at p. Prove the converse by producing a supporting functional for a hull vertex under the finite general-position hypotheses. ☐

Proof. **Detailed Learning Proof.**
TODO: Expand the two directions, making explicit where finiteness, supporting lines, and uniqueness from general position enter. ☐

Remark (Proof structure). The proof translates between extreme points, supporting lines, and strict maximizers of linear functionals.

Remark (Dependencies).

- Convex Hull
- Extreme Point
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- TODO: finite set maximum

Remark (Return). Return to Theorem

Remark (Proof vault). Handwritten proof record

Theorem (Monotone slope characterization of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ satisfy

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k is a vertex of

$$\operatorname{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Proof. **Professional Standard Proof.**
TODO: Prove that consecutive upper-hull edges must turn clockwise, which gives strictly decreasing slopes, and prove the converse by showing each listed point admits a supporting line. □

Proof. **Detailed Learning Proof.**
TODO: Expand the analytic-geometry argument for three ordered points and show how it applies along the whole upper chain. □

Remark (Proof structure). The proof is an ordered-slope argument: upper hull edges form a concave chain.

Remark (Dependencies).

- Convex Hull
- Supporting Line
- Supporting Functional Characterization of Hull Vertices
- TODO: slope of a line

Remark (Return). Return to Theorem

Remark (Proof vault). Handwritten proof record

Theorem (Greedy slope construction of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k with

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \arg\max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P, and the slopes encountered along the way are strictly decreasing.

Proof. **Professional Standard Proof.**
TODO: Show that each greedy slope choice selects the next upper-hull vertex, then use the monotone slope theorem to prove that no upper-hull vertex is skipped. □

Proof. **Detailed Learning Proof.**
TODO: Expand the induction on the produced vertices and identify where general-position uniqueness is used. □

Remark (Proof structure). The proof is the algorithmic consequence of the monotone slope characterization.

Remark (Dependencies).

- Convex Hull
- Monotone Slope Characterization of the Upper Hull
- TODO: finite set maximum
- TODO: slope of a line

Chapter 4

Computational Linear Algebra

Where You Are in the Journey

Algebraic Structures $\rightarrow$ Linear Algebra (Axler) $\rightarrow$ **Computational Linear Algebra** $\rightarrow$ Geometric Modeling $\rightarrow$ Differential Geometry $\rightarrow$ $\cdots$

Relationship to Axler. *Linear Algebra Done Right* is the theoretical spine for this material — vector spaces, linear maps, spectral theory, all developed rigorously and proof-first. This chapter is deliberately the other half: computational fluency with the same machinery. The two chapters are complementary and should not be conflated. Axler proves why the tools work; this chapter practises using them.

What this chapter builds. The computational toolkit needed to work through geometric modeling, numerical algorithms, and applied mathematics: vector arithmetic, matrix operations, solving linear systems, orthogonality, determinants, eigenvalues, the SVD, linear transformations, and selected applications. The structure follows Strang, *Introduction to Linear Algebra* (5th ed.), and sections are populated just-in-time as downstream work demands them.

Current priority. Chapters 1, 2 (ğ2.4–2.5), 3 (ğ3.4), 5 (ğ5.2), and 8 (ğ8.1–8.2) provide the linear algebra required for Piegl & Tiller, *The NURBS Book*, Chapters 1–3. The remaining sections are stubbed and will be populated as needed.

Structural Roadmap

$$\textbf{Computation} \longrightarrow \textbf{Geometry} \longrightarrow \textbf{Application}$$

The progression follows Strang's chapter sequence:

1. Vectors and linear combinations
2. Solving linear equations
3. Vector spaces and subspaces
4. Orthogonality
5. Determinants
6. Eigenvalues and eigenvectors
7. The Singular Value Decomposition
8. Linear transformations
9. Complex vectors and matrices
10. Applications
11. Numerical linear algebra
12. Linear algebra in probability and statistics

Remark (Primary source). Strang, *Introduction to Linear Algebra*, 5th ed. (2016); lectures freely available via MIT OpenCourseWare, 18.06.

4.1 Vectors and Linear Combinations

Vectors and Linear Combinations — Quick Reference

Vector in $\mathbb{R}^n$	An ordered list $(v_1, v_2, \ldots, v_n)$ of n real numbers.
Addition	$(v_1, \ldots, v_n) + (w_1, \ldots, w_n) = (v_1 + w_1, \ldots, v_n + w_n)$.
Scalar multiplication	$c(v_1, \ldots, v_n) = (cv_1, \ldots, cv_n)$.
Linear combination	$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$; scalars times vectors, summed.
Convex combination	Linear combination with $c_i \geq 0$ and $\sum c_i = 1$; result lies inside the convex hull.
Standard basis vectors	$\mathbf{e}_j \in \mathbb{R}^n$: all zeros except a 1 in position j. Every vector is a linear combination of $\mathbf{e}_1, \ldots, \mathbf{e}_n$.

4.1.1 Vectors

Definition (Vector in $\mathbb{R}^n$)

A *vector* in $\mathbb{R}^n$ is an ordered list of n real numbers, written $\mathbf{v} = (v_1, v_2, \ldots, v_n)$. The number n is the *dimension* of the vector.

Vector addition:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1,\ v_2 + w_2,\ \ldots,\ v_n + w_n).$$

Scalar multiplication:

$$c\mathbf{v} = (cv_1,\ cv_2,\ \ldots,\ cv_n), \quad c \in \mathbb{R}.$$

Remark (Geometric meaning). In $\mathbb{R}^2$ and $\mathbb{R}^3$, a vector is an arrow indicating displacement. Its entries specify length and direction, but *not* location the same vector can be placed anywhere in space. Addition corresponds to following one displacement and then another (head-to-tail rule). Scalar multiplicati stretches or shrinks the arrow; negative scalars reverse its direction.

Theorem (Properties of Vector Operations)

For all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$:

1. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity)
2. $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$ (associativity)
3. $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$ (distributivity over vector addition)
4. $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$ (distributivity over scalar addition)
5. $c(d\mathbf{v}) = (cd)\mathbf{v}$ (compatibility)

4.1.2 Linear Combinations

Definition (Linear Combination)

A *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ is any vector of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k,$$

where $c_1, c_2, \ldots, c_k \in \mathbb{R}$ are called the *coefficients*.
A *convex combination* is a linear combination in which $c_i \geq 0$ for all i and $\sum_{i=1}^{k} c_i = 1$.

Remark (Why convex combinations matter for NURBS). Every point on a Bézier or B-spline curve is a convex combination of control points. The Bernstein basis functions $B_{i,n}(u)$ are non-negative and sum to 1, so $C(u) = \sum_{i=0}^{n} B_{i,n}(u)P_i$ is a convex combination of the control points $\{P_i\}$ at every parameter value u. This is the *convex hull property*: the curve lies entirely within the convex hull of its control polygon.

Definition (Standard Basis Vectors)

For $j = 1, 2, \ldots, n$, the j-*th standard basis vector* $\mathbf{e}_j \in \mathbb{R}^n$ has a 1 in position j and 0s everywhere else. Every vector $\mathbf{v} = (v_1, \ldots, v_n) \in \mathbb{R}^n$ can be written as

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n.$$

4.2 Lengths and Dot Products

Lengths and Dot Products — Quick Reference

Dot product	$\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2 + \cdots + v_nw_n \in \mathbb{R}$.
Length (norm)	$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$.
Unit vector	$\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$; has length 1.
Angle	$\cos\theta = \dfrac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|}$.
Orthogonality	$\mathbf{v} \perp \mathbf{w}$ iff $\mathbf{v} \cdot \mathbf{w} = 0$.
Cauchy–Schwarz	$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\|\,\|\mathbf{w}\|$.

4.2.1 The Dot Product

Definition (Dot Product)

The *dot product* (or *inner product*) of $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is

$$\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2 + \cdots + v_nw_n \ \in \ \mathbb{R}.$$

Theorem (Properties of the Dot Product)

For all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

1. $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$ (commutativity)
2. $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$ (distributivity)
3. $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$ (scalar compatibility)
4. $\mathbf{v} \cdot \mathbf{v} \geq 0$, with equality iff $\mathbf{v} = \mathbf{0}$ (positive definiteness)

4.2.2 Length and Angle

Definition (Length and Unit Vector)

The *length* (or *norm*) of $\mathbf{v} \in \mathbb{R}^n$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

A *unit vector* has length 1. The *normalisation* of a nonzero vector $\mathbf{v}$ is $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$, which is the unit vector in the direction of $\mathbf{v}$.

Theorem (Cauchy–Schwarz Inequality)

For all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \, \|\mathbf{w}\|,$$

with equality if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel (one is a scalar multiple of the other).

Definition (Angle Between Vectors)

The *angle* $\theta \in [0, \pi]$ between nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is defined by

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|}.$$

The vectors are *orthogonal* (written $\mathbf{v} \perp \mathbf{w}$) if $\theta = \pi/2$, equivalently if $\mathbf{v} \cdot \mathbf{w} = 0$.

Remark (Dot product and normals in NURBS). The unit normal to a parametric surface $S(u, v)$ is computed as $N = S_u \times S_v / \|S_u \times S_v\|$, which requires both the cross product and normalisation via the norm. The dot product also enters the convex hull argument: showing that a B-spline curve lies on a specific side of a hyperplane reduces to checking the sign of dot products of control points with the hyperplane's normal.

4.3 Introduction to Matrices

Matrices — Quick Reference

Matrix	Rectangular array of numbers; $A \in \mathbb{R}^{m\times n}$ has m rows, n columns.
Matrix–vector	$(A\mathbf{x})_i = \sum_j A_{ij}x_j$; $A\mathbf{x}$ is a linear combination of columns of A.
Matrix–matrix	$(AB)_{ij} = \sum_k A_{ik}B_{kj}$; requires inner dimensions to match.
Transpose	$(A^T)_{ij} = A_{ji}$; rows become columns.
Identity	I_n: ones on diagonal, zeros elsewhere; $IA = AI = A$.
Column view	$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$ where $\mathbf{a}_j$ are the columns of A.

4.3.1 Matrices and Matrix-Vector Multiplication

Definition (Matrix)

An $m \times n$ *matrix* A is a rectangular array of real numbers with m rows and n columns:

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix}.$$

The entry in row i, column j is denoted A_{ij} or $[A]_{ij}$. We write $A \in \mathbb{R}^{m\times n}$.

Definition (Matrix–Vector and Matrix–Matrix Multiplication)

If $A \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^n$, then $A\mathbf{x} \in \mathbb{R}^m$ is defined by

$$(A\mathbf{x})_i = \sum_{j=1}^{n} A_{ij}x_j, \qquad i = 1, \dots, m.$$

Equivalently, $A\mathbf{x}$ is the linear combination of the columns of A with coefficients from $\mathbf{x}$:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, their product $AB \in \mathbb{R}^{m\times p}$ has entries

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj}.$$

Remark (Matrix multiplication is not commutative). In general $AB \neq BA$, even when both products are defined. Order matters: applying transformation B first, then A, is written AB.

4.3.2 The Transpose

Definition (Transpose)

The *transpose* of $A \in \mathbb{R}^{m\times n}$ is the matrix $A^T \in \mathbb{R}^{n\times m}$ defined by $(A^T)_{ij} = A_{ji}$. Rows of A become columns of A^T. Key properties:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T\mathbf{w}$ (dot product as matrix multiplication)

4.3.3 Matrix Form of a Curve

Remark (NURBS connection: matrix form of curves and surfaces). The power basis curve $C(u) = \sum_{i=0}^{n} \mathbf{a}_i u^i$ can be written as

$$C(u) = [\mathbf{a}_0 \;\; \mathbf{a}_1 \;\; \cdots \;\; \mathbf{a}_n] \begin{pmatrix} 1 \\ u \\ \vdots \\ u^n \end{pmatrix} = A\,\mathbf{u}(u),$$

where A is the matrix of coefficient vectors and $\mathbf{u}(u)$ is the monomial vector. A Bézier curve has the equivalent matrix form $C(u) = [u^i]^T M [P_i]$, where M is the $(n+1)\times(n+1)$ change-of-basis matrix converting Bernstein basis to power basis. A tensor product surface takes the form $S(u,v) = [f_i(u)]^T[b_{i,j}][g_j(v)]$, a matrix sandwiched between two basis-function vectors. All of these representations rely on matrix-vector multiplication as defined above.

4.4 Rules for Matrix Operations

Matrix Operations — Quick Reference

Associativity	$A(BC) = (AB)C$.
Distributivity	$A(B + C) = AB + AC$; $(A + B)C = AC + BC$.
NOT commutative	$AB \neq BA$ in general.
Identity	$AI = IA = A$.
Transpose product	$(AB)^T = B^T A^T$.
Diagonal matrices	$D_1 D_2$ is diagonal; DA scales rows; AD scales columns.

4.4.1 Algebraic Rules

Theorem (Rules for Matrix Multiplication)

When the dimensions are compatible:

1. $A(BC) = (AB)C$ (associativity)
2. $A(B + C) = AB + AC$ (left distributivity)
3. $(A + B)C = AC + BC$ (right distributivity)
4. $c(AB) = (cA)B = A(cB)$ for any scalar c
5. $(AB)^T = B^T A^T$ (transpose reverses order)

The following do **not** hold in general:

- $AB \neq BA$ (not commutative)
- $AB = AC \not\Rightarrow B = C$ (no cancellation in general)
- $AB = 0 \not\Rightarrow A = 0$ or $B = 0$

4.4.2 Diagonal Matrices

Definition (Diagonal Matrix)

A *diagonal matrix* $D \in \mathbb{R}^{n \times n}$ has $D_{ij} = 0$ for all $i \neq j$. Written $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$.
Key computational facts:

- DA multiplies each *row* i of A by d_i.
- AD multiplies each *column* j of A by d_j.
- $D_1 D_2 = \operatorname{diag}(d_1 e_1, \ldots, d_n e_n)$ where e_j are the diagonal entries of D_2.

4.4.3 Block Matrices

Remark (Block matrix multiplication). Matrices can be partitioned into blocks, and multiplied block-by-block as if the blocks were scalar entries, provided the block sizes are compatible. If $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ and $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$, then

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix},$$

provided inner block dimensions match. The standard matrix of a linear transformation $[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)]$ is a block matrix with column vectors as blocks, and $[T]\mathbf{v} = v_1 T(\mathbf{e}_1) + \cdots + v_n T(\mathbf{e}_n)$ is block matrix-vector multiplication.

4.5 Inverse Matrices

Inverse Matrices — Quick Reference

Inverse	A^{-1}: the unique matrix with $AA^{-1} = A^{-1}A = I$.
Existence	A^{-1} exists iff A is square and $\det(A) \neq 0$ (nonsingular).
2×2 formula	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
Product rule	$(AB)^{-1} = B^{-1}A^{-1}$ (order reverses).
Transpose rule	$(A^T)^{-1} = (A^{-1})^T$.
Change of basis	If M converts basis B_1 to B_2, then M^{-1} converts B_2 to B_1.

4.5.1 Definition and Properties

Definition (Invertible Matrix)

A square matrix $A \in \mathbb{R}^{n \times n}$ is *invertible* (or *nonsingular*) if there exists a matrix $A^{-1} \in \mathbb{R}^{n \times n}$ such that

$$AA^{-1} = A^{-1}A = I_n.$$

The matrix A^{-1} is the *inverse* of A and is unique. A matrix with no inverse is called *singular*.

Theorem (Properties of Inverses)

When A and B are invertible $n \times n$ matrices:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$ (order reverses)
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(cA)^{-1} = \frac{1}{c}A^{-1}$ for $c \neq 0$

4.5.2 The 2×2 Case

Formula (2×2 Inverse)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad ad - bc \neq 0.$$

The quantity $ad - bc$ is the determinant. The matrix is singular when $ad = bc$.

4.5.3 Change of Basis via Inverse

Remark (NURBS connection: converting between Bernstein and power basis). In Piegl & Tiller Exercise 1.14, the Bernstein basis functions of degree n can be written as $[B_{0,n}(u), \ldots, B_{n,n}(u)] = [1, u, \ldots, u^n]M$, where M is an $(n+1) \times (n+1)$ matrix. This means a Bézier curve has the matrix form

$$C(u) = [u^i]^T M [P_i],$$

where $[u^i]^T$ is the monomial row vector, $[P_i]$ is the column of control points, and M encodes the change from power basis to Bernstein basis.

The inverse M^{-1} converts the other direction: if you have a curve in power basis form with coefficients $[\mathbf{a}_i]$, then the equivalent Bézier control points are $[P_i] = M^{-1}[\mathbf{a}_i]$.

This is a direct application of the matrix inverse: M converts one basis to another, and M^{-1} inverts the conversion.

4.6 Independence, Basis, and Dimension

Basis and Dimension — Quick Reference

Linearly independent	No vector is a linear combination of the others; only $\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ has $c_i = 0$ for all i.
Span	$\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$: all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_k$.
Basis	A linearly independent set that spans the space.
Dimension	Number of vectors in any basis; all bases have the same size.
Change of basis	Transition matrix P: columns are the new basis vectors in terms of the old basis.

4.6.1 Linear Independence

Definition (Linear Independence)

Vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ are *linearly independent* if the only solution to

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

is $c_1 = c_2 = \cdots = c_k = 0$. Otherwise they are *linearly dependent*: at least one vector is a linear combination of the others.

4.6.2 Basis and Dimension

Definition (Basis)

A *basis* for a subspace $V \subseteq \mathbb{R}^n$ is a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ that is:

1. linearly independent, and
2. spans V (every vector in V is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$).

Theorem (Dimension is Well-Defined)

Any two bases for the same subspace V have the same number of vectors. This common number is called the *dimension* of V, written $\dim V$.

Remark. The standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$, so $\dim \mathbb{R}^n = n$. The key insight: every set of n linearly independent vectors in $\mathbb{R}^n$ is automatically a basis.

4.6.3 Change of Basis

Definition (Transition Matrix)

Let $\mathcal{B}_1 = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ and $\mathcal{B}_2 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be two ordered bases for $\mathbb{R}^n$. The *transition matrix* P from $\mathcal{B}_1$ to $\mathcal{B}_2$ has as its i-th column the coordinates of $\mathbf{u}_i$ expressed in the basis $\mathcal{B}_2$. If $[\mathbf{x}]_{\mathcal{B}_1}$ denotes the coordinate vector of $\mathbf{x}$ in basis $\mathcal{B}_1$, then

$$[\mathbf{x}]_{\mathcal{B}_2} = P\,[\mathbf{x}]_{\mathcal{B}_1}.$$

The inverse P^{-1} converts in the opposite direction.

Remark (NURBS connection: Bernstein basis as a basis for polynomials). The Bernstein polynomials $\{B_{0,n}(u), B_{1,n}(u), \dots, B_{n,n}(u)\}$ form a basis for the vector space of polynomials of degree $\leq n$. So does the monomial (power) basis $\{1, u, u^2, \dots, u^n$ The matrix M from Piegl & Tiller Exercise 1.14 is precisely the transition matrix converting coordinates in the monomial basis to coordinates in the Bernstein basis. M^{-1} converts the other direction. This is why the Bézier curve $C(u) = [u^i]^T M[P_i]$ can be read as: "express the Bernstein basis functions in the monomial basis via M, then multiply by the control points."

4.7 Determinants and the Cross Product

Determinants and Cross Product — Quick Reference

2×2 **det**	$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$
3×3 **det**	Cofactor expansion along any row or column.
Cross product	$\mathbf{v} \times \mathbf{w}$: vector orthogonal to both $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$, computed via 3×3 determinant mnemonic.
Magnitude	$\\|\mathbf{v}\times\mathbf{w}\\| = \\|\mathbf{v}\\|\,\\|\mathbf{w}\\| \sin\theta$; equals area of parallelogram.
Direction	Right-hand rule; $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$.

4.7.1 The 3×3 Determinant

Definition (3×3 Determinant via Cofactor Expansion)

For $A \in \mathbb{R}^{3\times 3}$, expanding along the first row:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

The signs alternate: $+, -, +$ along the first row.

4.7.2 The Cross Product

Definition (Cross Product)

For $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ in $\mathbb{R}^3$, the *cross product* is

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - v_3w_2,\;\; v_3w_1 - v_1w_3,\;\; v_1w_2 - v_2w_1).$$

Theorem (Properties of the Cross Product)

For $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$ and $c \in \mathbb{R}$:

1. $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$ (anticommutativity)
2. $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$ (distributivity)
3. $\mathbf{v} \times \mathbf{v} = \mathbf{0}$
4. $(c\mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$
5. $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ and $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ (orthogonality)
6. $\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\|\,\|\mathbf{w}\| \sin\theta$ (magnitude = parallelogram area)

Remark (NURBS connection: surface normal vectors). The unit normal to a parametric surface at a point (u,v) is

$$N(u,v) = \frac{S_u \times S_v}{\|S_u \times S_v\|},$$

where S_u and S_v are the partial derivative vectors of the surface with respect to the parameters. The cross product gives the direction orthogonal to both tangent directions; dividing by the norm produces the unit normal. This appears in Piegl & Tiller ğ1.1 and is used throughout whenever normal vectors are needed (shading, offset surfaces, etc.).

Remark (Mnemonic for the cross product). Write the entries of $\mathbf{v}$ and $\mathbf{w}$ each twice in a 2×6 array. Ignore the first and last columns. Draw three "X" shapes between columns 23, 34, and 45. Each "X" contributes one entry of $\mathbf{v}\times\mathbf{w}$: add along forward diagonals, subtract along backward diagonals.

4.8 The Idea of a Linear Transformation

Linear Transformations — Quick Reference

Definition	$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w}$ and scalars c, d.
Origin fixed	Always $T(\mathbf{0}) = \mathbf{0}$; translation is *not* linear.
Standard matrix	$[T] = [T(\mathbf{e}_1) \mid \cdots \mid T(\mathbf{e}_n)]$; then $T(\mathbf{v}) = [T]\mathbf{v}$.
Composition	$S \circ T$ has matrix $[S][T]$.
Affine map	$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$: linear part A plus translation $\mathbf{v}$.
Invariance	B-spline curves: apply affine map to control points only.

4.8.1 Definition

Definition (Linear Transformation)

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a *linear transformation* if

$$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and all $c, d \in \mathbb{R}$.
Equivalently, T must satisfy both:

1. $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ (preserves addition), and
2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (preserves scalar multiplication).

Remark (The origin is always fixed). Every linear transformation satisfies $T(\mathbf{0}) = \mathbf{0}$. Therefore, *translation* (shifting by a nonzero vector) is not linear. The four geometric operations that *are* linear: rotation about the origin, scaling, reflection, and projection.

4.8.2 Geometric Catalog in $\mathbb{R}^2$

Standard Planar Transformations

Scaling	$\begin{pmatrix} c_1 & 0 \\ 0 & c_2 \end{pmatrix}$: scales x by c_1, y by c_2.
Rotation by θ	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$: rotates counter-clockwise.
Reflection ($y = x$)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: swaps x and y coordinates.
Projection (x-axis)	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$: drops the y component.

Remark (Composition is matrix multiplication). Applying T_1 then T_2 has matrix $[T_2][T_1]$. A sequence of geometric operations on a model (scale, then rotate, then translate) corresponds to multiplying their matrices together in that order.

4.8.3 Affine Transformations

Definition (Affine Transformation)

An *affine transformation* $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ has the form

$$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v},$$

where $A \in \mathbb{R}^{n\times n}$ is the linear part and $\mathbf{v} \in \mathbb{R}^n$ is a translation vector. An affine transformation is a linear transformation when $\mathbf{v} = \mathbf{0}$.

Remark (Affine invariance of B-spline curves NURBS Property P3.4). An affine transformation $\Phi(\mathbf{r}) = A\mathbf{r}+\mathbf{v}$ applied to a B-spline curve $C(u) = \sum_{i=0}^{n} N_{i,p}(u)P_i$ gives

$$\Phi(C(u)) = \sum_{i=0}^{n} N_{i,p}(u)\,\Phi(P_i).$$

The transformation can be applied to the curve by applying it to each control point individually. This works because the basis functions sum to 1 (partition of unity):

$$\Phi(C(u)) = A\Big(\sum N_{i,p}P_i\Big) + \mathbf{v} = \sum N_{i,p}(AP_i) + \underbrace{\Big(\sum N_{i,p}\Big)}_{=1}\mathbf{v} = \sum N_{i,p}(AP_i + \mathbf{v}).$$

This means rotating, translating, or scaling a B-spline curve costs only $n+1$ affine operations on control points, regardless of how many points are evaluated on the curve.

4.9 The Matrix of a Linear Transformation

Standard Matrix — Quick Reference

Standard matrix	$[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \in \mathbb{R}^{m\times n}$.
How to build it	Evaluate T on each standard basis vector; place results as columns.
Uniqueness	$[T]$ is the *unique* matrix with $T(\mathbf{v}) = [T]\mathbf{v}$ for all $\mathbf{v}$.
Composition	$[S \circ T] = [S][T]$.
Change of basis	Same T, different bases $\Rightarrow$ different matrix; related by $B = P^{-1}AP$.

4.9.1 The Standard Matrix

Theorem (Standard Matrix of a Linear Transformation)

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there exists a unique matrix $[T] \in \mathbb{R}^{m\times n}$ such that

$$T(\mathbf{v}) = [T]\mathbf{v} \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

The *standard matrix* $[T]$ is constructed by placing the images of the standard basis vectors as columns:

$$[T] = \big[\, T(\mathbf{e}_1) \;\big|\; T(\mathbf{e}_2) \;\big|\; \cdots \;\big|\; T(\mathbf{e}_n) \,\big].$$

Remark (Why this works). Since every $\mathbf{v} = v_1\mathbf{e}_1 + \cdots + v_n\mathbf{e}_n$, linearity gives

$$T(\mathbf{v}) = v_1T(\mathbf{e}_1) + \cdots + v_nT(\mathbf{e}_n) = [T]\mathbf{v}.$$

Knowing T on the basis vectors determines T on the entire space.

4.9.2 Matrix Under Change of Basis

Definition (Matrix Relative to a Basis)

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation and let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be an ordered basis. The *matrix of T relative to $\mathcal{B}$*, denoted $[T]_{\mathcal{B}}$, has as its j-th column the coordinates of $T(\mathbf{b}_j)$ expressed in the basis $\mathcal{B}$.
If P is the transition matrix whose columns are $\mathbf{b}_1, \dots, \mathbf{b}_n$ expressed in the standard basis, then

$$[T]_{\mathcal{B}} = P^{-1}[T]P.$$

Two matrices related by $B = P^{-1}AP$ are called *similar*.

Remark (NURBS connection: Bézier curve in matrix form). The Bézier curve has the matrix representation

$$C(u) = [u^i]^T M[P_i],$$

where:

- $[u^i]^T = (1, u, u^2, \dots, u^n)$ is the monomial basis vector,
- $[P_i]$ is the column vector of control points,
- M is the $(n+1)\times(n+1)$ matrix that encodes the Bernstein polynomials in the monomial basis.

The matrix M is the standard matrix of the linear transformation that converts from Bernstein basis coordinates to monomial basis coordinates. Setting $[\mathbf{a}_i] = M[P_i]$ gives the power basis representation of the same curve (so the Bézier coefficients $[P_i]$ are the coordinates of the curve in the Bernstein basis, and $[\mathbf{a}_i]$ are the coordinates in the monomial basis). Converting between the two is exactly the change-of-basis operation $P^{-1}AP$ applied to the coefficient vectors.

Exercises

Exercises: Computational Linear Algebra

Vectors and Linear Combinations

1. Compute the following:
   (a) $(2, -1, 3) + (-1, 4, -2)$
   (b) $3(1, 0, -2) - 2(-1, 1, 1)$
   (c) $\|(3, -4)\|$, $\|(1, 2, 2)\|$, and $\|(0, 0, 0, 5)\|$
2. Write $(5, -3, 7)$ as a linear combination of the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$.
3. Determine whether $(1, 2, 3)$ is a linear combination of $(1, 1, 0)$ and $(0, 1, 1)$. If yes, find the coefficients. If no, explain why not.
4. The control points of a quadratic Bézier curve are $P_0 = (0, 0)$, $P_1 = (1, 2)$, $P_2 = (3, 0)$. The curve is $C(u) = (1-u)^2 P_0 + 2u(1-u)P_1 + u^2 P_2$.
   (a) Compute $C(0)$, $C(1/2)$, and $C(1)$.
   (b) Verify that $C(1/2)$ is a convex combination of P_0, P_1, P_2 (check that the weights are non-negative and sum to 1).
   (c) Verify that $C(1/2)$ lies inside the triangle formed by the control points.
5. Let $P_0 = (0, 0)$, $P_1 = (2, 3)$, $P_2 = (4, 0)$ be control points. The de Casteljau algorithm at $u = 1/3$ computes:

   $$P_0^1 = (1-u)P_0 + uP_1, \quad P_1^1 = (1-u)P_1 + uP_2, \quad C(u) = (1-u)P_0^1 + uP_1^1.$$

   Carry out this computation and find $C(1/3)$.

Dot Products and Lengths

6. Compute the dot product and the angle between the following pairs of vectors. State whether they are orthogonal, acute, or obtuse.
   (a) $\mathbf{v} = (1, 0, 0)$ and $\mathbf{w} = (0, 1, 0)$
   (b) $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (4, -2, 0)$
   (c) $\mathbf{v} = (1, 1)$ and $\mathbf{w} = (1, -1)$
7. Find the unit vector in the direction of $(3, -4, 0)$. Then find the unit vector in the direction of $(1, 1, 1)$.
8. The tangent vector to the quadratic Bézier curve at $u = 0$ is $C'(0) = 2(P_1 - P_0)$, and at $u = 1$ is $C'(1) = 2(P_2 - P_1)$. Using the control points $P_0 = (0, 0)$, $P_1 = (2, 1)$, $P_2 = (4, 0)$:
   (a) Compute $C'(0)$ and $C'(1)$.
   (b) Find the angle between the two endpoint tangent vectors.
   (c) Find the unit tangent vectors at each endpoint.

Matrices and Matrix Operations

9. Compute the following matrix products, or state why they are not defined.

   (a) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

   (b) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$

   (c) $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}$

10. The quadratic Bézier curve $C(u) = \sum_{i=0}^{2} B_{i,2}(u)P_i$ can be written in matrix form as

$$C(u) = \begin{pmatrix} 1 & u & u^2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} P_0 \\ P_1 \\ P_2 \end{pmatrix}.$$

    Using $P_0 = (0, 0)$, $P_1 = (1, 2)$, $P_2 = (2, 0)$, compute $C(1/2)$ by carrying out the matrix products in order from right to left.

11. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$. Compute AB, BA, and confirm that $AB \neq BA$. Also compute $(AB)^T$ and $B^T A^T$, and confirm they are equal.

Inverse Matrices and Change of Basis

12. Find the inverse of each matrix, or state that it is singular.

    (a) $\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$

    (b) $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$

    (c) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

13. For the linear Bézier case ($n = 1$), the conversion matrix between the Bernstein basis $\{1 - u, u\}$ and the monomial basis $\{1, u\}$ is

$$M = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

    (a) Verify that $[B_{0,1}(u),\, B_{1,1}(u)] = [1, u]\, M$, i.e., that $(1 - u) = 1 \cdot 1 + u \cdot (-1)$ and $u = 1 \cdot 0 + u \cdot 1$.

    (b) Find M^{-1}.

    (c) The power basis curve $C(u) = \mathbf{a}_0 + \mathbf{a}_1 u$ has $\mathbf{a}_0 = (1, 0)$, $\mathbf{a}_1 = (2, 1)$. Using $[P_i] = M^{-1}[\mathbf{a}_i]$, find the equivalent Bézier control points.

Independence and Basis

14. Determine whether each set of vectors is linearly independent. If dependent, express one vector as a combination of the others.

    (a) $\{(1,2),(3,6)\}$

    (b) $\{(1,0,1),(0,1,1),(1,1,0)\}$

    (c) $\{(1,1,0),(0,1,1),(1,0,-1)\}$

15. The Bernstein polynomials of degree 2 are $B_{0,2}(u) = (1-u)^2$, $B_{1,2}(u) = 2u(1-u)$, $B_{2,2}(u) = u^2$.

    (a) Write each as a linear combination of $\{1, u, u^2\}$.

    (b) Verify that $B_{0,2} + B_{1,2} + B_{2,2} = 1$ (partition of unity).

    (c) Write the 3×3 matrix M whose rows are the monomial coefficients of $B_{0,2}$, $B_{1,2}$, $B_{2,2}$.

Cross Product and Surface Normals

16. Compute the cross product $\mathbf{v} \times \mathbf{w}$ and verify orthogonality by checking $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$.

    (a) $\mathbf{v} = (1,0,0)$, $\mathbf{w} = (0,1,0)$

    (b) $\mathbf{v} = (1,2,3)$, $\mathbf{w} = (-1,0,1)$

    (c) $\mathbf{v} = (2,-1,0)$, $\mathbf{w} = (0,3,-1)$

17. A parametric surface patch has $S_u = (1,0,2)$ and $S_v = (0,1,-1)$ at a point.

    (a) Compute $S_u \times S_v$.

    (b) Find the unit normal vector $N = (S_u \times S_v)/\|S_u \times S_v\|$.

    (c) Find the area of the parallelogram spanned by S_u and S_v.

Linear and Affine Transformations

18. For each matrix A, describe the geometric effect of the transformation $T(\mathbf{x}) = A\mathbf{x}$ on $\mathbb{R}^2$, and compute $T(1,0)$ and $T(0,1)$.

    (a) $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

    (b) $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

    (c) $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

19. Find the standard matrix of the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ that rotates vectors counter-clockwise by 45. *Hint: find $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ using trigonometry.*

20. The affine transformation $\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$ has

$$A = \begin{pmatrix} \cos 30 & -\sin 30 \\ \sin 30 & \cos 30 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

The quadratic Bézier curve has control points $P_0 = (0,0)$, $P_1 = (1,1)$, $P_2 = (2,0)$. By affine invariance, the transformed curve has control points $\Phi(P_0)$, $\Phi(P_1)$, $\Phi(P_2)$. Compute the three new control points.

21. A 3-point homogeneous control point is $\tilde{P} = (w \cdot x, w \cdot y, w)$.

    (a) For $P = (2,3)$ with weight $w = 2$, write the homogeneous point $\tilde{P}$.

    (b) Apply the perspective map H to recover P from $\tilde{P}$.

    (c) For $\tilde{P}_0 = (1,0,1)$, $\tilde{P}_1 = (1,1,1)$, $\tilde{P}_2 = (0,2,2)$ (the homogeneous circular arc from Piegl & Tiller Example 1.13), compute the de Casteljau midpoint $\tilde{C}^w(1/2)$ using the formula $\tilde{C}^w = (1-u)^2\tilde{P}_0 + 2u(1-u)\tilde{P}_1 + u^2\tilde{P}_2$ at $u = 1/2$, then project via H to find $C(1/2)$.

Chapter 5

Computational Probability

Status: Planned

Active mathematical content has not yet been added.

Computational Probability Notes

This chapter is planned. Active mathematical content has not yet been added.

Proofs

Proof entries for this chapter are planned. No proof files have been regenerated.

Exercises

Exercises for this chapter are planned. No exercise content has been restored here yet.

Chapter 6

Ordinary Differential Equations

Where This Chapter Lives

Calculus → **Ordinary Differential Equations** → Dynamical Systems / Numerical Methods / PDE

Why this chapter belongs here. Ordinary differential equations are where derivatives stop being only local rates and start governing whole families of evolving states. Once differentiation is in hand, equations such as

$$y'(t) = f(t, y(t))$$

become the natural language for motion, growth, decay, forcing, and feedback.

What this chapter will gather. The eventual ODE chapter is the right home for:

- first-order equations and separable models;
- linear equations and integrating factors;
- existence and uniqueness ideas;
- systems of ODEs and phase portraits;
- stability, equilibrium, and qualitative behavior;
- numerical methods such as Euler and Runge–Kutta schemes.

Why this is currently a breadcrumb. The surrounding curriculum is still building the supporting architecture: calculus, linear algebra, and numerical analysis. ODEs sit at the meeting point of all three, so this file marks the destination clearly even before the chapter is fully written.

6.1 Models

Definition 6.1.1 (Differential Equation). A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.

For an unknown function $y = y(t)$, a differential equation may be written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0.$$

Here t is the independent variable, $y(t)$ is the unknown function, and

$$y', y'', \ldots, y^{(n)}$$

are derivatives of y.

Remark (Interpretation). At the computational level, a differential equation describes how a quantity changes. Instead of giving the function $y(t)$ directly, it gives a relation involving the rate of change of $y(t)$, and possibly higher-order rates of change.

Example 6.1.2 (A First Differential Equation). The equation

$$y' = 3y$$

is a differential equation because it relates the unknown function y to its derivative y'.

A solution is

$$y(t) = Ce^{3t},$$

where C is a constant. Indeed,

$$y'(t) = 3Ce^{3t} = 3y(t).$$

Definition 6.1.3 (Order of a Differential Equation). The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.

For example, if an equation involving an unknown function $y = y(t)$ can be written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0$$

and $y^{(n)}$ is the highest derivative that appears, then the differential equation has **order** n.

Example 6.1.4 (Orders of Differential Equations). The equation

$$y' = 3y$$

is first order, since the highest derivative appearing is y'.

The equation

$$y'' + 4y' + 5y = 0$$

is second order, since the highest derivative appearing is y''.

The equation

$$y^{(4)} - ty'' + y = 0$$

is fourth order, since the highest derivative appearing is $y^{(4)}$.

Definition 6.1.5 (Solution of a Differential Equation). A **solution** of a differential equation is a function that satisfies the equation on a specified interval.

More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0,$$

then a solution on an interval I is a function $y : I \to \mathbb{R}$ such that y has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \ldots, y^{(n)}(t)) = 0$$

for every $t \in I$.

Remark (Interpretation). A solution is not a number; it is a function. The function solves the differential equation when substituting the function and its derivatives into the equation makes the equation true for every point in the interval under consideration.

Example 6.1.6 (Checking a Solution). The function

$$y(t) = Ce^{3t}$$

is a solution of

$$y' = 3y$$

on $\mathbb{R}$, because

$$y'(t) = 3Ce^{3t} = 3y(t)$$

for every $t \in \mathbb{R}$.

Definition 6.1.7 (Autonomous Differential Equation). An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.

For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of y depends only on the current value of y, not directly on time t.

Remark (Interpretation). An autonomous differential equation describes a system whose rule of evolution does not change with time. If the system is in the same state y at two different times, then the differential equation assigns the same rate of change at both times.

Example 6.1.8 (Autonomous Equations). The equation

$$y' = 3y$$

is autonomous, since the right-hand side depends only on y.

The equation

$$y' = t + y$$

is not autonomous, since the independent variable t appears explicitly.

Definition 6.1.9 (Non-Autonomous Differential Equation). A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.

For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of y may depend both on the current value of y and on the independent variable t.

Remark (Interpretation). A non-autonomous differential equation describes a system whose rule of evolution may change over time. Even if the system has the same state y at two different times, the differential equation may assign different rates of change because the value of t may be different.

Example 6.1.10 (Non-Autonomous Equations). The equation

$$y' = t + y$$

is non-autonomous, since the independent variable t appears explicitly.

The equation

$$y' = 3y$$

is autonomous, not non-autonomous, since the right-hand side depends only on y.

Definition 6.1.11 (Linear Differential Equation). A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.

An nth-order linear ordinary differential equation for an unknown function $y = y(t)$ has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

and the forcing term $g(t)$ are given functions of the independent variable t.

Remark (Interpretation). Linearity means that the equation is linear in the unknown function y and its derivatives. The coefficients may depend on t, but they may not depend on y or its derivatives. Thus terms such as y^2, yy', $\sin(y)$, and $(y')^2$ make an equation nonlinear.

Example 6.1.12 (Linear and Nonlinear Equations). The equation

$$y'' + 3ty' + 2y = \sin t$$

is linear.

The equation

$$y'' + y^2 = 0$$

is nonlinear because it contains the term y^2.

The equation

$$y' + yy' = t$$

is nonlinear because it contains the product yy'.

Ordinary differential equations are often used to model physical phenomena because many natural systems are governed by relationships between a quantity and its rate of change. In such models, the unknown function represents the state of the system, while the differential equation describes how that state evolves over time. For example, population models describe how a population changes, cooling models describe how temperature changes, motion models describe how position and velocity change, and circuit models describe how current or voltage changes. In each case, the differential equation does not usually give the quantity directly; instead, it gives a rule for how the quantity changes.

Definition 6.1.13 (Malthusian Population Growth Model). The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.

If $P(t)$ denotes the population at time t, then the model is

$$P'(t) = kP(t),$$

where k is a constant called the **growth rate** or **proportionality constant**.

Remark (Interpretation). The model says that population growth depends only on the current population size. If $k > 0$, the population grows exponentially. If $k < 0$, the population decays exponentially. If $k = 0$, the population remains constant.

Example 6.1.14 (General Solution). The general solution of

$$P'(t) = kP(t)$$

is

$$P(t) = Ce^{kt},$$

where C is a constant. If the initial population is $P(0) = P_0$, then

$$P(t) = P_0 e^{kt}.$$

Definition 6.1.15 (Decay). A quantity is said to undergo **decay** if it decreases over time.

In a differential equation model, if $Q(t)$ denotes the quantity at time t, then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where $k > 0$ is a constant.

Remark (Interpretation). The equation

$$Q'(t) = -kQ(t)$$

says that the rate of change of the quantity is proportional to the current amount, but with a negative sign. Thus larger amounts decay faster in absolute size, while smaller amounts decay more slowly.

Example 6.1.16 (Exponential Decay). The general solution of

$$Q'(t) = -kQ(t)$$

is

$$Q(t) = Ce^{-kt}.$$

If $Q(0) = Q_0$, then

$$Q(t) = Q_0 e^{-kt}.$$

For $k > 0$, this function decreases toward 0 as t increases.

Definition 6.1.17 (Newton's Law of Cooling). **Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.

If $T(t)$ denotes the temperature of the object at time t, and T_a denotes the ambient temperature, then the model is

$$T'(t) = -k\big(T(t) - T_a\big),$$

where $k > 0$ is a constant called the **cooling constant**.

Remark (Interpretation). The quantity

$$T(t) - T_a$$

measures how far the object's temperature is from the ambient temperature. Newton's Law of Cooling says that the larger this temperature difference is, the faster the object cools or warms toward the ambient temperature.

Example 6.1.18 (General Solution). The general solution of

$$T'(t) = -k\big(T(t) - T_a\big)$$

is

$$T(t) = T_a + Ce^{-kt},$$

where C is a constant. If $T(0) = T_0$, then

$$T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

Definition 6.1.19 (General Solution). A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.

For an nth-order ordinary differential equation, the general solution typically contains n arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n).$$

Each choice of the constants $C_1, C_2, \ldots, C_n$ gives a particular solution of the differential equation.

Remark (Interpretation). The general solution represents the full family of solutions produced by solving the differential equation. Initial conditions are then used to choose specific values of the arbitrary constants, producing a particular solution.

Example 6.1.20 (General Solution). The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t},$$

where C is an arbitrary constant. If an initial condition such as

$$y(0) = 5$$

is imposed, then $C = 5$, and the particular solution is

$$y(t) = 5e^{3t}.$$

Definition 6.1.21 (Family of Solutions). A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters.

For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \dots, C_n),$$

where $C, C_1, \dots, C_n$ are parameters. Each choice of the parameters gives one particular solution.

Remark (Interpretation). A family of solutions records many solutions at once. The parameters act as labels that distinguish one solution curve from another. When initial conditions are supplied, they usually determine the parameter values and select a single member of the family.

Example 6.1.22 (A Family of Solutions). The differential equation

$$y' = 3y$$

has the family of solutions

$$y(t) = Ce^{3t},$$

where $C \in \mathbb{R}$. Each value of C gives a different solution.

Definition 6.1.23 (Initial Condition). An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.

For a first-order differential equation involving an unknown function $y = y(t)$, an initial condition has the form

$$y(t_0) = y_0,$$

where t_0 is the initial time and y_0 is the initial value.

Remark (Interpretation). An initial condition selects one solution from a family of solutions. If a differential equation has general solution

$$y(t) = \Phi(t, C),$$

then the condition

$$y(t_0) = y_0$$

is used to determine the constant C.

Example 6.1.24 (Initial Condition). The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t}.$$

If the initial condition is

$$y(0) = 5,$$

then

$$5 = Ce^0 = C.$$

Thus the particular solution satisfying the initial condition is

$$y(t) = 5e^{3t}.$$

Definition 6.1.25 (Initial Value Problem). An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.

For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time t_0.

Remark (Interpretation). An initial value problem asks for the particular solution of a differential equation that passes through a specified point. Geometrically, the condition

$$y(t_0) = y_0$$

means that the solution curve must pass through the point

$$(t_0, y_0).$$

Example 6.1.26 (Initial Value Problem). Consider the initial value problem

$$y' = 3y, \qquad y(0) = 5.$$

The differential equation has general solution

$$y(t) = Ce^{3t}.$$

Using the initial condition,

$$5 = y(0) = Ce^0 = C.$$

Therefore the solution of the initial value problem is

$$y(t) = 5e^{3t}.$$

Definition 6.1.27 (Existence). An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.

For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

existence means that there is a function $y(t)$, defined on some interval containing t_0, such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

Definition 6.1.28 (Uniqueness). An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.

For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

uniqueness means that if $y_1(t)$ and $y_2(t)$ are both solutions on the same interval I containing t_0, then

$$y_1(t) = y_2(t)$$

for every $t \in I$.

Definition 6.1.29 (Interval of Existence). An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.

For an initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

an interval I is an interval of existence if $t_0 \in I$ and there is a solution $y : I \to \mathbb{R}$ such that

$$y'(t) = f(t, y(t))$$

for every $t \in I$, and

$$y(t_0) = y_0.$$

Remark (Interpretation). Existence asks whether at least one solution passes through the initial point. Uniqueness asks whether only one solution passes through that point. The interval of existence records how long the solution is valid.

Example 6.1.30 (Existence and Uniqueness). The initial value problem

$$y' = 3y, \qquad y(0) = 5$$

has the solution

$$y(t) = 5e^{3t}.$$

This solution exists and is unique on $\mathbb{R}$.

The interval of existence is

$$(-\infty, \infty).$$

Proofs

To be populated.

Capstone

To be populated.

Chapter 7

Partial Differential Equations

Status: Planned

Active mathematical content has not yet been added.

Partial Differential Equations Notes

This chapter is planned. Active mathematical content has not yet been added.

Proofs

Proof entries for this chapter are planned. No proof files have been regenerated.

Exercises

Exercises for this chapter are planned. No exercise content has been restored here yet.

Chapter 8

Fourier Analysis

fourier-analysis

Calculus $\rightarrow$ Differential Equations $\rightarrow$ **Fourier Analysis** $\rightarrow$ $\cdots$

Proofs

To be populated.

Capstone

To be populated.

Bibliography